ECE 2020: Fundamentals of Digital System Design

Causey, Section B, Fall 2025

Homework #5: Due October 8

1. Convert from one base to another. Show all your work explicitly.

   (a) 1011100.101_2 (2's complement) to decimal

   (b) 1011100.101_2 (unsigned) to decimal

   (c) -45_{10} to 2's complement binary

   (d) -45_{10} to sign-magnitude hexadecimal

   (e) 52_{10} to hexadecimal

   (f) 52_8 to hexadecimal

   (g) 10110101_2 (unsigned) to ternary (base 3)

2. Show how the following calculations are done in 1's complement 8-bit binary numbers. Do not change the order of operands.

   (a) $-31 + 64$

   (b) $-65 - 30$

   (c) $31 - 65$

3. Show how the following calculations are done in 2's complement 8-bit binary numbers. Do not change the order of operands.

   (a) $-25 + 48$

   (b) $-35 - 49$

   (c) $34 - 49$

1

a $1011100.101_2 \rightarrow 0100011010_2 + 1$

$= 0100011011_2 = 2^0 + 2^1 + 2^2 + 2^3 + 2^7 = 283$

$283/2^3 = 35.375$ $\boxed{-35.375}$

b 1011100.101

$= 2^6 + 2^4 + 2^3 + 2^2 + 2^{-1} + 2^{-3} = \boxed{92.625}$

c -45_{10}

$45_{10} - 2^5 = 13$ $\quad$ 00101101

$13 - 2^3 = 5$ $\quad$ $\downarrow$

$5 - 2^2 = 1$ $\quad$ $11010010 + 1$

$1 - 2^0 = 0$ $\quad$ $= \boxed{11010011_2}$

d $-45_{10} = 00101101_2 \rightarrow 10101101_2$

$1010 = A$

$1101 = D$

$\boxed{AD}$

e 52_{10}

$\boxed{34}$

$3 \cdot 16^1 = 48$
$4 \cdot 16^0 = + 4$
52

f $52_8 = 5 \cdot 8^1 + 2 \cdot 8^0$
$40 + 2 = 42$

$\boxed{2A}$

$16^1 \cdot 2 = 32 +$
$16^0 \cdot A = 10$
42

g $10110101_2 =$

$1 \cdot 2^0 = 1$
$1 \cdot 2^2 = 4$
$1 \cdot 2^4 = 16 +$
$1 \cdot 2^5 = 32$
$1 \cdot 2^7 = 128$
181

$181 \div 3 = 60 \text{ r } 1$
$60 \div 3 = 20 \text{ r } 0$
$20 \div 3 = 6 \text{ r } 2$
$6 \div 3 = 2 \text{ r } 0$
$2 \div 3 = 0 \text{ r } 2$

$\boxed{20201_3}$

2

a -31 $+64$

00011111 01000000

11100000

$$\begin{array}{r} 11100000 \\ +\ 01000000 \\ \hline 1100100000 \end{array}$$

$\to 00100001_2$

$2^0 \cdot 1 + 2^5 \cdot 1 = \boxed{33}$

b $65 = 2^6 + 2^0$

$01000001 \to 10111110$

$30 = 2^4 + 2^3 + 2^2 + 2^1 \to 00011110 \to 11100001$

$$\begin{array}{r} 10111110 \\ +\ 11100001 \\ \hline 110011111 \end{array}$$

$\to 10100000 \to 01011111$

$2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^6$
$= \boxed{-95}$

c

$31 = 00011111$ +
$-65 = 10111110$

$11011101 \rightarrow 00100010$

$32 + 2 = 34 = \boxed{-34}$

3

a $-25 \rightarrow 2^4 \cdot 1 + 2^3 \cdot 1 + 2^0 \cdot 1$

$00011001 \rightarrow 11100110 + 1 = 11100111$

$46 \rightarrow 2^5 \cdot 1 + 2^4 \cdot 1 \rightarrow 00110000$

11001111

$\begin{array}{r} 11100111 \\ +\ 11001111 \\ \hline 00010111 \end{array}$

$2^0 + 2^1 + 2^2 + 2^4 = \boxed{23}$

b $-35 \Rightarrow 2^5 + 2^1 + 2^0 \Rightarrow 00100011$

$\downarrow$

11011100

$+ \quad 1$

11011101

$49 \Rightarrow 2^5 + 2^4 + 2^0 \Rightarrow 00110001$

$\downarrow$

11001110

$+ \quad 1$

11001111

11011101

11001111

$10101100 \Rightarrow 01010011$

$2^7 + 2^5 + 2^1 + 2^0 = \boxed{-84}$

c $34 = 2^5 + 2^1 = 00100010$

$-49 = 2^5 + 2^4 + 2^0 = 00110001$

$\downarrow$

$11001110 + 1$

$= 11001111$

111

11001111

$+ \; 00100010$

$11110001 \Rightarrow 00001110 + 1 = 00001111$

$2^4 + 2^3 + 2^2 + 2^1 = \boxed{-15}$